

1. Tech Stack

Frontend: React (Vite + JSX)

- Component-based architecture for clean code separation
- Fast and modern development experience with Vite
- Ecosystem rich in UI libraries, routing, state management

Backend: Firebase (Firestore + Auth)

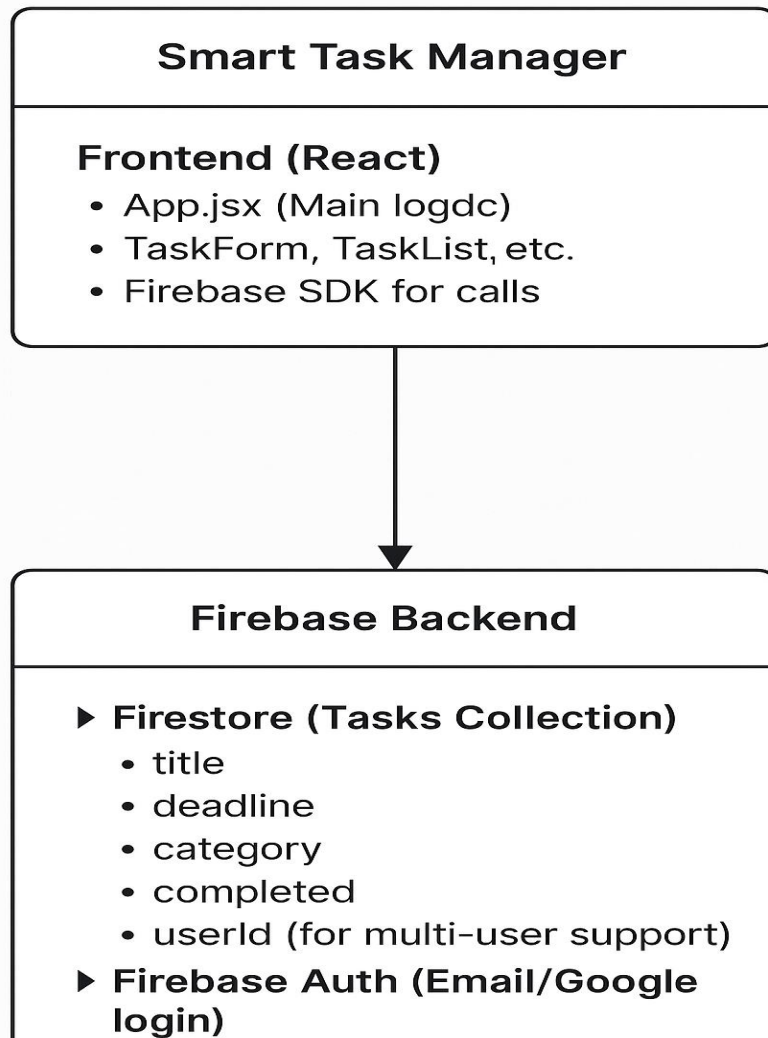
- No backend server setup needed (BaaS)
- **Realtime database sync**
- Built-in **Authentication (Google, email/password)**
- Scalable and secure

State Management: React `useState` + Custom Hooks

2. Why This Stack?

Tool	Why Chosen
React	Fast, modular UI, ideal for task apps and I have more experience in working with React.
Vite	Faster dev builds than CRA
Firebase	Handles authentication and real-time sync out of the box
Firestore	NoSQL database that scales well and works seamlessly with React & Firebase
Dayjs	Lightweight date formatting and comparison

3. High-Level Architecture Diagram



4. Database Schema (Firestore - NoSQL)

Collection: `tasks`

Each task document:

```
{  
  "title": "string",  
  "category": "string",  
  "deadline": "YYYY-MM-DD",  
  "completed": "boolean",  
  "userId": "string (auth UID)"  
}
```

Sample:

```
{  
  "id": "auto-generated",  
  "title": "Finish report",  
  "category": "Work",  
  "deadline": "2025-07-06",  
  "completed": false,  
  "userId": "abc123" // links to Firebase Auth user  
}
```